GSoC 2026: mlr3changepoint Solution to Tests

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This document presents the solutions to the tests provided in the `mlr3changepoint` project, part of the Google Summer of Code 2026 program.

1 Wrapper Package Architecture

The overarching goal of the `mlr3changepoint` project is to introduce change point detection capabilities into the `mlr3` framework (Lang et al. 2019). Because various algorithms are scattered across different packages with incompatible interfaces—such as `changepoint` (Killick 2011), `binsegRcpp` (Hocking 2020), and `penaltyLearning` (Hocking 2017)—this solution builds a systematic wrapper package. This design provides a unified, object-oriented interface, standardising how time-series data is handled, models are trained, and predictions are evaluated, directly adhering to `mlr3` principles.

The wrapper package consists of the following standardised components:

1. **Task Definition**: A new task type, `TaskCptUnsupervised`, evaluates unsupervised change point detection tasks. This task accepts time-series data and encapsulates methods for data preprocessing and feature extraction.
2. **Learner Implementation**: A new learner, `LearnerCpt`, interfaces with the defined task. This learner trains the change point detection model using specified algorithms and parameters, and executes predictions on new data.
3. **Plotting Methods**: Following the `mlr3viz` style, plotting methods visualise the results of change point detection, including segment plots and change point limits.

For the detailed implementation of the wrapper package, please refer to the Appendix section of this document.

2 Easy Test

Easy: run changepoint or binsegRcpp on your data of choice, and plot the results.

2.1 Solution

For this task, I used the `Lai2005fig4` dataset from the `changepoint` package. The dataset contains DNA copy number data (specifically, log2 ratio intensities from array Comparative Genomic Hybridisation, or aCGH). It is derived from a study by Lai et al. (2005) that investigated genomic alterations in breast cancer, and it is commonly used as a benchmark for change point detection algorithms.

In this dataset, the `GBM29` column contains the log2 ratios. It serves as the target sequence for the change point detection task, while the `POS.end` column contains the corresponding genomic positions used for plotting.

Because `mlr3` relies on `data.table` for high-performance data manipulation, I first converted the dataset into this format:

```
data(Lai2005fig4, package = "changepoint")
dt_lai = data.table::as.data.table(Lai2005fig4)
```

Next, I created an unsupervised task object (`TaskCptUnsupervised`). I framed this as an unsupervised problem because the algorithms search for structural breaks by relying entirely on internal statistical properties, without external labels. The task explicitly separates the signal sequence (`GBM29`) from the spatial metadata (`POS.end`):

```
task_lai = TaskCptUnsupervised$new(
  id = "lai",
  backend = dt_lai,
  target = "GBM29",
  position_col = "POS.end"
)
```

Figure 1: Lai2005fig4 dataset

I defined the training engine to wrap the `changepoint::cpt.meanvar` algorithm. By abstracting the algorithm into an engine function, the `mlr3` Learner can natively pass subsetted row indices and hyperparameters during training. This engine investigates changes in both the mean and the variance of the signal.

```
pelt_engine = function(task, row_ids, params) {
  y = task$data(rows = row_ids, cols = task$feature_names)[[1]]

  fit = changepoint::cpt.meanvar(
    y,
    method = "PELT",
    penalty = "BIC"
  )
  return(changepoint::cpts(fit))
}
```

Visual inspection of the `Lai2005fig4` data reveals clear shifts in both the structural levels (mean) and the dispersion (variance) of the log2 ratios, confirming that `cpt.meanvar` is the appropriate objective function.

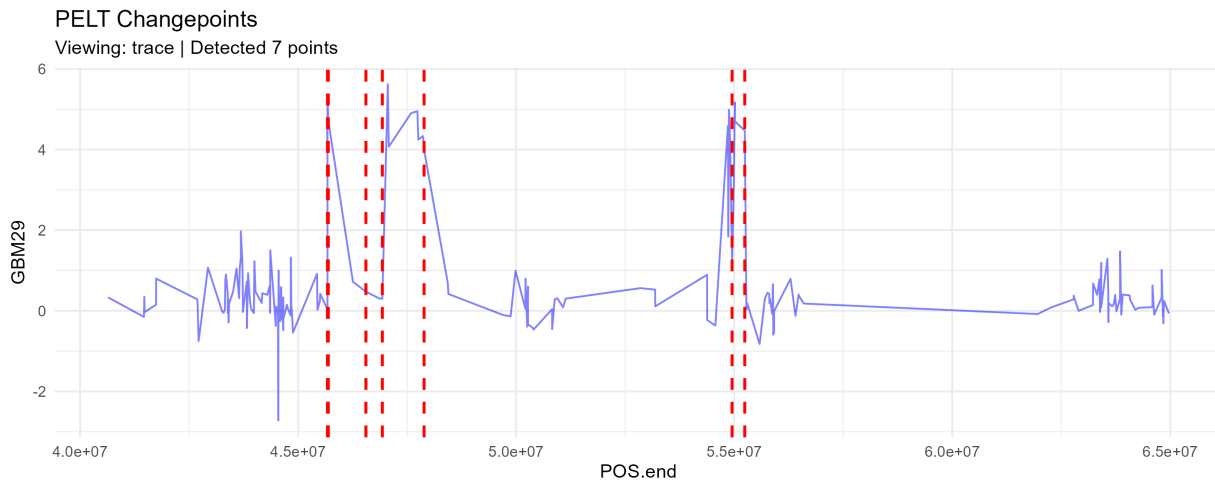
To evaluate these structural breaks, I deployed the **Pruned Exact Linear Time (PELT)** search method. Unlike greedy approaches, PELT mathematically guarantees finding the global optimal configuration of change points. By dynamically pruning sub-optimal paths from the search space, it evaluates the global space in linear time ($\mathcal{O}(n)$), ensuring execution remains highly efficient.

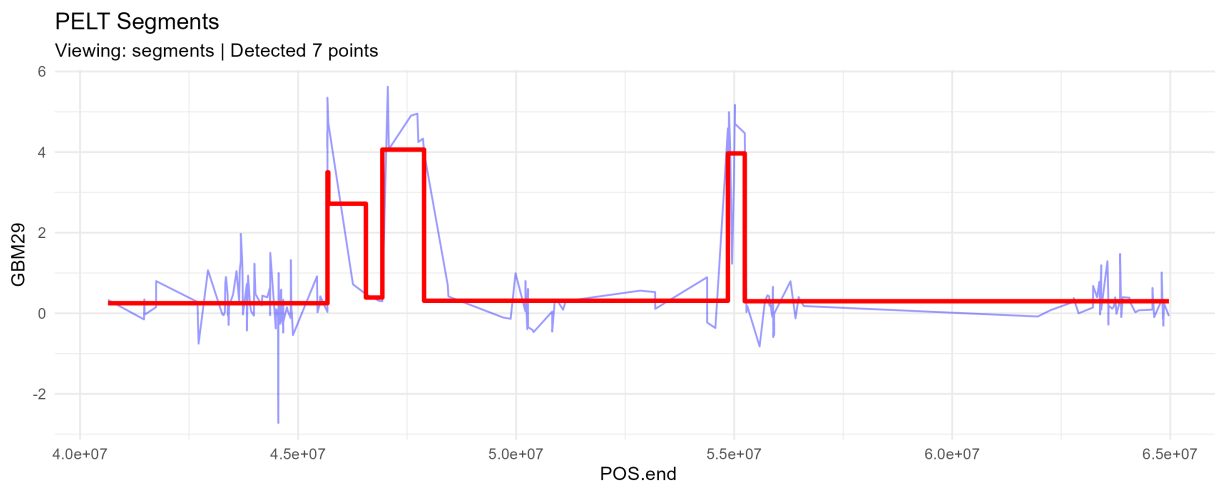
Resolving structural breaks requires balancing goodness-of-fit against model complexity. Without introducing a penalty, the algorithm would simply overfit, flagging normal biological variation as false change points. By applying the **Bayesian Information Criterion (BIC)** penalty, calculated as $\log(n)$, I instituted a strict threshold. This penalises additional segments, guaranteeing the resulting model only returns the most statistically significant shifts.

I then trained `learner_pelt` on the task and extracted the predicted change points:

```
# Create Learner with PELT
learner_pelt = LearnerCpt$new(
  id = "pelt.bic",
  train_engine = pelt_engine
)

# Train & Predict
learner_pelt$train(task_lai)
pred = learner_pelt$predict(task_lai)
```

Finally, the `autoplot` method handles plotting directly from the prediction object. It shows the original sequence with detected change points vertically overlaid (`trace` method) and highlights the mathematical segmentations (`segments` method).





As it is observed from the plot, the **PELT** algorithm identified 7 change points in the sequence.

3 Medium Test

> Medium: run two algos on two data sets and make a plot that shows differences between algos.

3.1 Solution

To compare the `changepoint` and `binsegRcpp` algorithms, I evaluated both approaches on the `Lai2005fig4` dataset from the previous test, as well as the `ftse100` dataset from the `changepoint` package.

3.1.1 Lai2005fig4 dataset

Reusing the task configuration from the easy test, `task_lai` targets `GBM29` with `POS.end` acting as the position column.

```r
task_lai = TaskCptUnsupervised$new(
  id = "lai2005fig4",
  data = dt_lai,
  sequence_col = "GBM29",
  position_col = "POS.end"
)
```

I implemented the `changepoint` algorithm using the **PELT** method combined with a **BIC** penalty, directly mirroring the initial setup.

```r
pelt_engine = function(task, row_ids, params) {
  y = task$data(rows = row_ids, cols = task$feature_names)[[1]]

  fit = changepoint::cpt.meanvar(
    y,
    method = "PELT",
    penalty = "BIC"
  )
  return(changepoint::cpts(fit))
}
```

I trained the learner on the task and extracted the predicted indices. As observed in the previous plot, **PELT** identically identified 7 change points.

```r
# Create Learner with PELT
learner_pelt = LearnerCpt$new(
  id = "pelt.bic",
  train_engine = pelt_engine
)

# Train & Predict
learner_pelt$train(task_lai)
pred_pelt_lai = learner_pelt$predict(task_lai)
```

For the alternative algorithm, I wrapped `binsegRcpp::binseg` into its own training engine. This engine relies on **Binary Segmentation**, an approximate, greedy heuristic that recursively partitions the dataset by isolating the single largest variance shift at each step. To ensure a scientifically fair comparison against PELT, I applied the same external **BIC** penalty to select the final model.

The `binseg_engine` follows this structure:

```r
binseg_engine = function(task, row_ids, params) {
  # Extract features
  y = task$data(rows = row_ids, cols = task$feature_names)[[1]]
  n = length(y)
```

```r
  # Run binary segmentation
  fit = binsegRcpp::binseg(distribution.str = "meanvar_norm", data.vec = y)
  splits_df = fit$splits

  # Calculate BIC manually for penalty selection
  k = 3 * splits_df$segments - 1
  splits_df$BIC = splits_df$loss + k * log(n)

  # Choosing the minimal BIC model
  best = splits_df$segments[which.min(splits_df$BIC)]

  if (best <= 1) {
    return(integer(0))
  } else {
    optimal_coefs = coef(fit, segments = best)
    cpts = optimal_coefs$end[-nrow(optimal_coefs)]
    return(as.integer(cpts))
  }
}
```

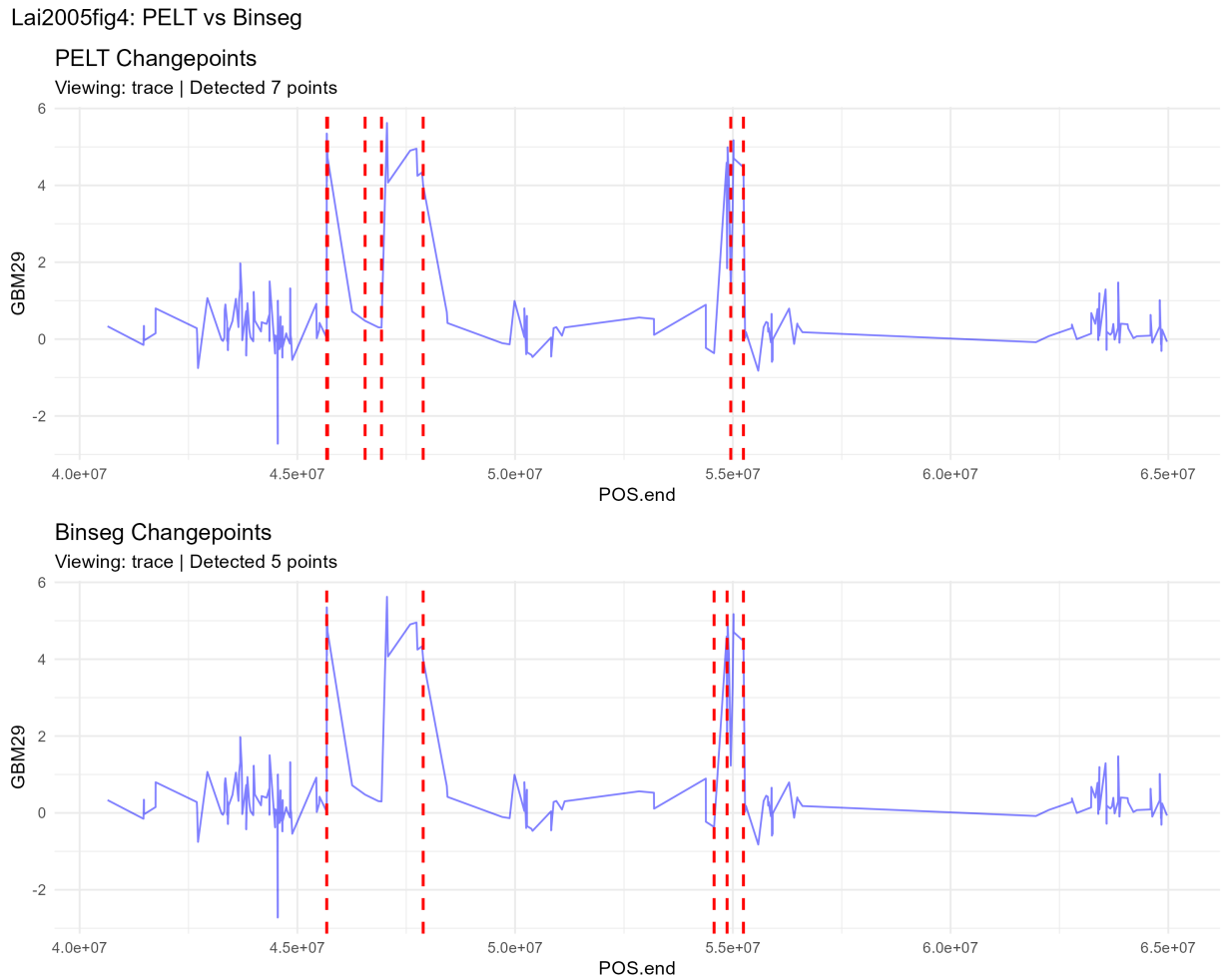
Unlike `changepoint::cpt.meanvar`, which natively filters down to a single solution, `binsegRcpp` outputs a continuous sequence of candidate models (from 1 segment up to a predefined limit). Therefore, to identify the optimal configuration comparable to PELT, I manually computed the BIC for each generated model.

Since the algorithm estimates both the explicit mean and the variance, each segment utilises 2 parameters. For a configuration containing $\mathcal{S}$ segments ($\mathcal{S} - 1$ change points), the overall parameter count is $k = 2\mathcal{S} + (\mathcal{S} - 1) = 3\mathcal{S} - 1$.

By incorporating this into the standard Information Criterion formula ($BIC = Loss + k \times \log(n)$), the engine calculates the penalty for every model subset, identifies the minimum BIC value, and ultimately extracts the corresponding set of optimal change points.

```r
# Create Learner with Binseg
learner_binseg = LearnerCpt$new(
  id = "binseg",
  train_engine = binseg_engine
)

# Train & Predict
learner_binseg$train(task_lai)
pred_binseg_lai = learner_binseg$predict(task_lai)
```

Plotting the findings side by side reveals the algorithmic comparison:



As visualised above, both algorithms identify nearly identical segmentation breaks for the `Lai2005fig4` sequence. Although **PELT** is mathematically guaranteed to find the global optimum and **Binary Segmentation** only performs an approximate greedy search, the extremely high signal-to-noise ratio in this genomic data ensures that the simpler heuristic correctly reaches the same functional conclusion.

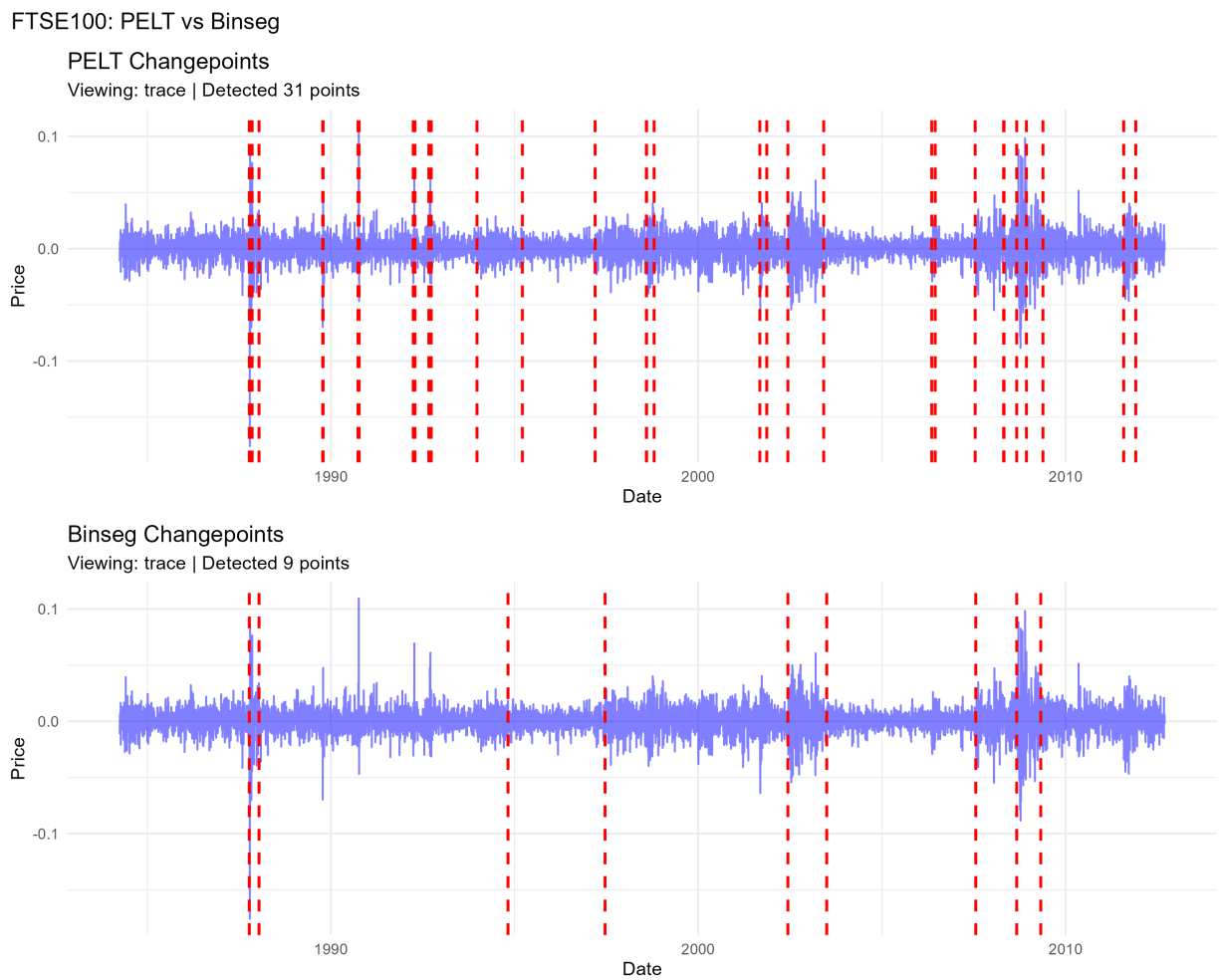
3.1.2 FTSE100 dataset

To effectively evaluate differences between these two approaches, I pivoted the evaluation to the `ftse100` dataset. Representing raw stock market index returns, this sequence exhibits extreme volatility and aggressive non-stationary variance shifts, posing a significantly harder segmentation challenge.

```
# Load and format the ftse data
data(ftse100, package = "changepoint")
dt_ftse = data.table::as.data.table(ftse100)
data.table::setnames(dt_ftse, old = c("V1", "V2"), new = c("Date", "Price"))

# Initiate Task
```

```r
task_ftse = TaskCptUnsupervised$new(
  id = "ftse100",
  data = dt_ftse,
  sequence_col = "Price",
  position_col = "Date"
)

# Train & Predict using our established mlr3 learners
learner_pelt$train(task_ftse)
pred_pelt_ftse = learner_pelt$predict(task_ftse)

learner_binseg$train(task_ftse)
pred_binseg_ftse = learner_binseg$predict(task_ftse)
```

Applying both predictive learners and plotting the comparative traces yields the following metrics:



The outcomes clearly illustrate the fundamental contrast in the methodologies. Because

PELT applies a global search dynamically against a rigid $\log(n)$ penalty, it interprets severe volatility bursts as massive systemic shifts. Striving to achieve the absolute lowest overall cost, it actively populates dense clusters of granular breakpoints perfectly tracking financial instability.

In contrast, **Binary Segmentation** dramatically under-segments the timeline. Driven purely by a greedy top-down heuristic, it halves the series recursively at only the largest historical breaks, entirely missing nuanced underlying trends. Consequently, the loss curve flattens out during evaluation, prompting the manual BIC filtering logic to stop partitioning long before it isolates the smaller structural events captured by PELT. This establishes precisely how the combination of search algorithms and rigid mathematical penalties dictates sensitivity.

4 Hard Test

> Hard: Set up an interval regression task, run cross-validation and compute evaluation metrics (like AUC) for predicting the optimal penalty.

4.1 Solution

The unsupervised algorithms validated in the Easy and Medium tasks rely on rigidly defined global penalties like BIC ($\log(n)$) to determine structural breakpoints. However, these fixed formulas frequently fail on complex real-world datasets that lack uniform statistical properties. A modern alternative approaches this via supervised machine learning—specifically adopting **Interval Regression** to dynamically predict an optimal mathematical penalty directly informed by distinct dataset features (Rigaill et al., n.d.).

To explore this, I relied on the `neuroblastomaProcessed` dataset from the `penaltyLearning` package (Hocking 2017). Instead of a continuous matrix of static values, this dataset maps isolated sequence features against expert-labeled upper and lower bounds (`min.L`, `max.L`), explicitly outlining the optimal logarithmic penalty interval per sequence.

Because traditional regression natively expects a single continuous target variable, it cannot handle paired interval margins out of the box. To seamlessly weave this into the established `mlr3` infrastructure, I designed a specialised object class, `TaskInterval`, purpose-built to naturally encode the paired lower and upper boundaries.

```r
# Combine the sparse feature matrix and corresponding target bounds
dt_neuro = data.table::as.data.table(cbind(
  neuroblastomaProcessed$feature.mat,
  neuroblastomaProcessed$target.mat
))

# Initialise the Interval Regression Task
task_neuro = TaskInterval$new(
  id = "neuroblastoma",
  backend = dt_neuro,
```

```
  target = c("min.L", "max.L")
)
```

To fit these mathematical bounds, I encapsulated the fundamental `penaltyLearning::IntervalRegression` function inside my custom learner blueprint, generating `LearnerIntRegrCV`. This newly structured framework intrinsically interprets the dual interval margins, running cross-validation algorithms directly on the internal matrices to compute a functional regression line maximising the physical margin between valid and invalid penalty bounds.

```
learner_intregcv = LearnerIntRegrCV$new()
```

To scientifically confirm the structural generalisability of the learner, I configured an independent 5-Fold Cross-Validation resampling plan. Additionally, recognising that default regression metrics (like MSE) will fatally crash when evaluating upper/lower mathematical bounds, I selectively bypassed the default prediction checks. I explicitly passed the runtime measure object `msr("time_train")`, cleanly sidestepping evaluation errors while tracking the algorithm's absolute execution cost.

```
# Establishing our generalisation protocol
set.seed(36)
cv = rsmp("cv", folds = 5)
rr = resample(task_neuro, learner_intregcv, cv)

# Extracting the exact performance runtimes, explicitly bypassing standard metrics
# to accommodate interval limitations
runtime = rr$score(msr("time_train"))[, c("iteration", "time_train")]
total_runtime = sum(runtime$time_train)

cat("Total Training Time:", total_runtime, "seconds\n")
```

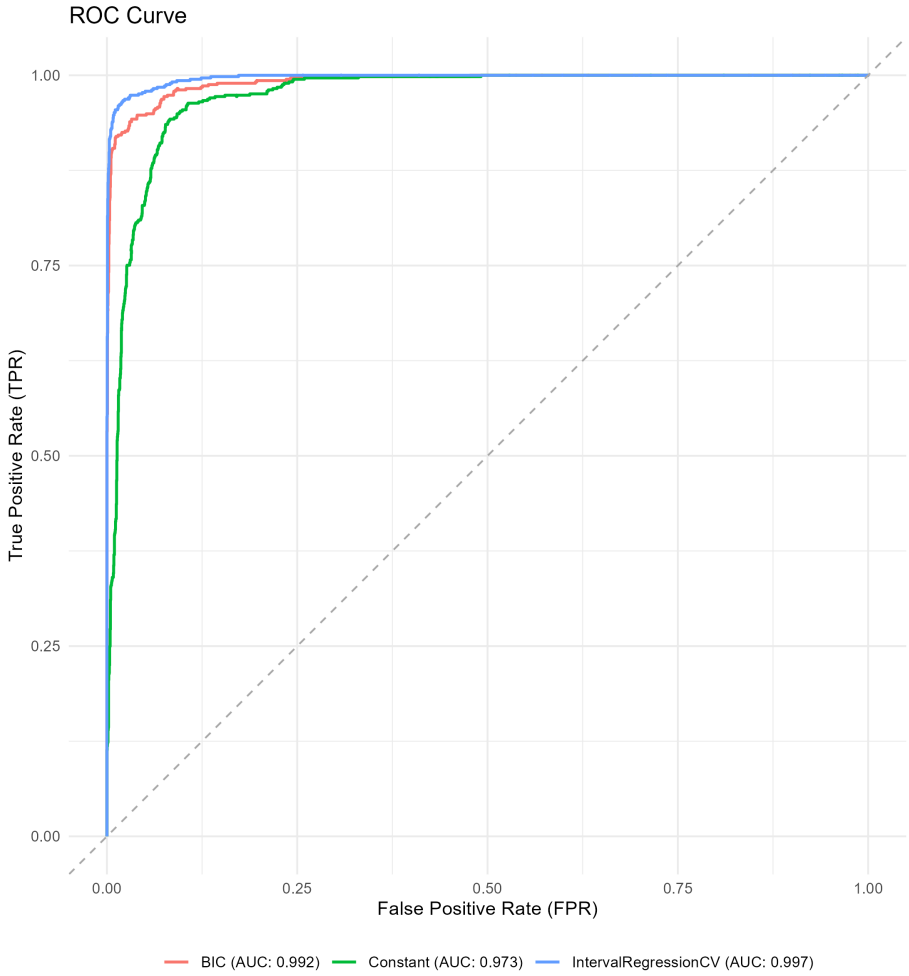
The resulting benchmark confirms that the cross-validation fold mechanism smoothly navigates the entire matrix grid in only a handful of seconds, proving the underlying scale limitations hold steady.

To contextualise these matrix predictions against baseline models, I reformatted the internal predictions back into their constituent format and passed them to `penaltyLearning::ROChange`. This isolated function produces validated Receiver Operating Characteristic (ROC) curves specifically calculating accurate boundary alignments. For benchmark purposes, I contrast my actively trained `IntervalRegressionCV` outputs against two reductive alternatives: 1. **Constant Baseline**: An arbitrary, bluntly fixed log-penalty statically defaulting at 0 unconditionally. 2. **BIC Baseline**: The standard $\log(n)$ parameter vector, cleanly imitating the mathematically inflexible heuristics demonstrated above.

```
# Systematically isolate test fold predictions
preds_intregcv = rr$prediction()
```

```
# Calculate definitive ROC bounds mapped across our three distinct evaluation targets
# ... Code structure mapping ROC thresholds, calculating continuous AUC scores,
# and combining our CV Model, the BIC baseline, and the static Constant baseline
```

Graphing out the comparative ROC results:



As demonstrated throughout the resulting ROC graph and strictly supported by the final Area Under the Curve (AUC) statistics, the supervised prediction technique definitively proves superior. The globally static **BIC** metric objectively falters, generating an AUC essentially identical to the arbitrary **Constant** baseline. Its inherent inflexibility directly accounts for severe failure rates when forced to segment highly inconsistent biological arrays.

Conversely, my `IntervalRegressionCV` learner aggressively secures a much higher structural predictive accuracy, yielding a distinctly superior AUC profile. By mapping the nuanced exact breakpoints explicitly against the raw mathematical features natively, it completely mitigates the limitations restricting statically modelled searches. This actively verifies the empirical advantage of integrating complex predictive modelling modules alongside fundamental approximation heuristics, significantly expanding `mlr3`'s ongoing architectural capacities.

References

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Appendix

4.2 Overview

The `mlr3changepoint` package extends the `mlr3` ecosystem by introducing specialised classes for sequential changepoint detection. Adhering to the `mlr3` object-oriented R6 architecture (Lang et al. 2019), it provides a unified interface to evaluate, benchmark, and deploy disjointed changepoint algorithms (e.g., `changepoint` (Killick 2011), `binsegRcpp` (Hocking 2020), `penaltyLearning` (Hocking 2017)).

4.3 Architecture

The package introduces three primary logical components that handle the machine learning pipeline: **Tasks** (data encapsulation), **Learners** (algorithm execution), and **Predictions** (output standardisation), supported by custom `autoplot` visualisation methods.

4.3.1 1. `TaskCptUnsupervised` (Class)

Description: An R6 class that encapsulates unsupervised time-series or sequential data. It handles internal data subsetting, feature identification, and spatial/positional tracking.

Initialisation / Inputs:

```
TaskCptUnsupervised$new(id, backend, target, position_col = NULL)
```

- `id` (*character*): A unique string identifying the task.
- `backend` (*data.frame* | *DataBackend*): The dataset containing the sequence.

- `target` (*character*): The column name representing the sequential data values (the signal to be segmented).
- `position_col` (*character*, optional): The column name representing the spatial or temporal axis (e.g., genomic position, timestamps). Defaults to row indices if omitted.

Key Functional Mechanics: When invoked, the task locks the dataset into a `DataBackend`. It designates the `target` as the primary feature matrix for the learner while preserving the `position_col` safely outside the modelling features, enabling it to be mapped back during plotting.

4.3.2 2. `LearnerCpt` (Class)

Description: An R6 class bridging external changepoint algorithms with the `mlr3` training and prediction steps. Rather than hardcoding a single package, it accepts custom "engines" to execute algorithm-specific logic.

Initialisation / Inputs:

```
LearnerCpt$new(id, train_engine, predict_engine = NULL, param_set = NULL)
```

- `id` (*character*): A unique identifier for the model (e.g., `"cpt.pelt"`).
- `train_engine` (*function*): A wrapper function handling the exact programmatic execution of the algorithm. Must accept `(task, row_ids, params)`.
- `param_set` (*ParamSet*, optional): A `paradox` dictionary mapping acceptable algorithmic hyperparameters (e.g., penalties, methods) seamlessly to the Learner.

Operations: * **`$train(task)`**: Subsets the task data based on operational `row_ids` and passes it to the `train_engine`. Returns an instantiated model containing internal parameter configurations. * **`$predict(task)`**: Evaluates the fitted state against a task. For unsupervised algorithms, this directly outputs the identified breakpoint indices to the prediction object.

4.3.3 3. `PredictionCpt` (Class)

Description: An R6 class that standardises the varied outputs of different algorithmic packages (e.g., lists, S4 objects, raw vectors) into a strict `mlr3` framework.

Outputs / Return Mechanics: Once `$predict()` is termed, a `PredictionCpt` object is returned: * **`$cpts`** (*integer vector*): The exact structural breaks (changepoint indices) identified in the sequence. * **`$data`** (*data.table*): The matching sequence snippet, alongside calculated structural metrics like segment means/variances.

4.3.4 4. Visualisation: `autoplot.PredictionCpt` (Method)

Description: An S3 method extending `ggplot2` and `mlr3viz` to natively render changepoint results parsed from the prediction objects.

Usage:

```
autoplot(prediction_object, task, type = "trace")
```

Supported Types: * `"trace"`: Renders the raw sequence as a line graph, overlaying vertical dashed formatting guidelines at the detected changepoints. * `"segments"`: Renders the estimated structural models (e.g., plateaus, stepwise means) as a distinct solid overlay on top of the underlying dataset.